

Tim G Williams

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<https://tgwilliams.com/>

Citizenship: New Zealand; United Kingdom; US Green Card Holder
Languages: English (native); German (B1)
Current location: Amherst, MA, USA

Summary

- 10 years' experience analyzing sustainable agri-food systems, incl. carbon credits and climate resilience
- Excellent quantitative data science skills: GIS, simulation modelling, risk analysis, machine learning
- Passionate about bridging social equity and environmental sustainability
- Big thinker who enjoys finding opportunities for strategic innovation
- Highly focused and competent at working with both people and data
- Hobbies include plant-based cooking and fermentation, trail running, watercolor, and cycle touring

Education

2021 *Ph.D. in Industrial and Operations Engineering*, University of Michigan, USA
Thesis title: Agent-Based Modeling of Resilience in Smallholder Agriculture: Toward Robust Models and Equitable Outcomes

2019 *M.Sc. in Industrial and Operations Engineering*, University of Michigan, USA

2015 *B.Eng. in Natural Resources Engineering*, University of Canterbury, New Zealand

Work experience

2025- **Senior Scientist**
Ecological-Economic Policy Modelling Group, University of Hohenheim | Germany

- Modeled ecological-economic feedbacks in organic agricultural systems, including coupling a land-use model with a process-based model of pollinator abundance
- Reviewed the state-of-the-art approaches for linking models of the economy to biodiversity
- Assisted with grant applications and group management
- Supervised and mentored MSc and PhD students

2024-25 **Environmental Modeler & Researcher**
Climate Farmers | Germany

- Developed and implemented scientific improvements for a carbon credit methodology, focusing on modeling soil carbon sequestration and GHG emissions
- Created data validation protocols, helping to scale from 6 to 38 farms and improve communication between internal stakeholders and farmers
- Implemented a Microsoft Excel version of the company's GHG model, which enhanced transparency and received glowing reviews from a third-party auditor
- Developed a dynamic remote sensing index, enabling a more scalable and realistic representation of regenerative farming impacts
- Collaborated extensively in cross-functional teams (e.g., science, product, tech)

2021-24 **Postdoctoral Researcher**
Institute for Environmental Studies, Vrije Universiteit Amsterdam | Netherlands

- Led 3 publications on data-driven analysis of power dynamics in European food systems
- Supervised 4 MSc students in successfully conducting and defending their theses
- Gave a keynote presentation at an international conference and led a workshop at a food system summit for citizens and activists
- Co-led a working group for the department's diversity, equity, and inclusion initiative

2016-21 **Ph.D. Candidate**
Industrial and Operations Engineering, University of Michigan | USA

- Exceptionally high productivity: published 7 first-author peer reviewed articles
- Modelled scenarios for mitigating climate risk in smallholder agricultural systems
- Utilized GIS analysis to quantify land-use change and inequalities in urban accessibility

- Proposals and funding acquisition: received prestigious Rackham Predoctoral Fellowship
- Contributed actively to university life: co-led a sustainability research community for 4 years, leading bi-weekly workshops and organizing annual conferences

2016

Graduate Environmental Engineer

Auckland Water Team, Beca Ltd | New Zealand

- Engineering design and report writing for water treatment plants, pipeline routing, and landfill leachate tanks

Technical Skills

Programming	Experienced in Python and R, and a fast learner
Data analysis	Extensive experience applying quantitative methods (simulation, statistics, machine learning, geospatial analysis (GIS)) to different types of data (household surveys, remote sensing, agricultural censuses). Git version control. Parallel computation.
Software	Microsoft Office, QGIS, LaTeX, ATLAS.ti, Github

Generalizable skills

Analysis	The bedrock of my education and practice is rigorous, quantitative analysis. I am versed in a wide array of analysis techniques, and my research has generated advances in simulation modelling and land-use change analysis.
Project management	My research background has given me experience in planning and executing complex and highly uncertain projects. At Climate Farmers I helped coordinate tasks across cross-functional teams, involving (re)evaluating timelines and division of responsibility.
Writing	I have published 9 first-author peer reviewed articles in interdisciplinary journals and written successful grant applications.
Proactivity	I am an active member of my professional communities. For instance, I have founded two university clubs – UC Meditation and ARRO (Adventure Racing, Rogaining, and Orienteering) – demonstrating innovation and a drive to foster community.
Independence	I have thrived in independent, remote environments, with a track record of self-discipline, initiative, and productivity in virtual teams.
Teamwork	My colleagues compliment my ability to bring together people working on different aspects of a project. I am most happy when collaborating on ideas, analysis, and writing.

Selected Peer-Reviewed Publications

“Power dynamics shape sustainability transitions in a modeled food system.” Williams et al. (2025) [One Earth](#)

“Synthesising the diversity of European agri-food networks: A meta-study of actors and power-laden interactions.” Williams et al. (2023) [Global Environmental Change](#)

“Resilience and equity: Quantifying the distributional effects of resilience-enhancing strategies in a smallholder agricultural system.” Williams et al. (2020) [Agricultural Systems](#)

“Parks and safety: A comparative study of green space access and its inequities in five US cities.” Williams et al. (2020) [Landscape and Urban Planning](#)

All other publications at [Google Scholar](#)

Key Strengths

Gallup Strengths Finder® strengths:

Harmony	I believe that little is gained from conflict and aspire to unite people via their similarities.
Achiever	I have a constant need for attainment
Relator	I derive pleasure and strength from forming close relationships with people.
Includer	I want to include people and make them feel a part of the group. I am instinctively accepting of others, regardless of race, gender identity, nationality, personality, or faith.

Learner I constantly strive to learn and improve. The process of learning is as important as the knowledge I gain.